SEA·BIRD SCIENTIFIC

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SENSOR SERIAL NUMBER: 9567
CALIBRATION DATE: 10-Apr-25

Slocum Payload CTD PRESSURE CALIBRATION DATA
1450 psia S/N 11201507

COEFFICIENTS:

PA0 =	1.031005e-02	PTCA0 =	5.238990e+05
PA1 =	4.522734e-03	PTCA1 =	4.243440e-02
PA2 =	-2.317772e-11	PTCA2 =	9.674988e-03
PTEMPA0 =	-6.190734e+01	PTCB0 =	2.512275e+01
PTEMPA1 =	5.424346e-02	PTCB1 =	9.500000e-04
PTEMPA2 =	-7.199889e-07	PTCB2 =	0.000000e+00

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.49	527141.4	1584.6	14.64	0.01
301.63	590623.6	1586.6	301.40	-0.02
588.83	654294.2	1587.8	588.83	0.00
876.00	717963.6	1588.2	876.07	0.01
1163.05	781627.2	1589.0	1163.09	0.00
1450.18	845322.8	1589.0	1450.07	-0.01
1163.20	781660.6	1588.8	1163.24	0.00
876.15	718003.0	1588.6	876.25	0.01
589.01	654340.2	1588.6	589.04	0.00
301.75	590661.8	1588.8	301.58	-0.01
14.49	527123.8	1591.6	14.56	0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1783	527207.00
29.00	1715	527216.20
24.00	1619	527207.70
18.50	1512	527186.00
15.00	1446	527209.60
4.50	1245	527211.00
1.00	1178	527190.70

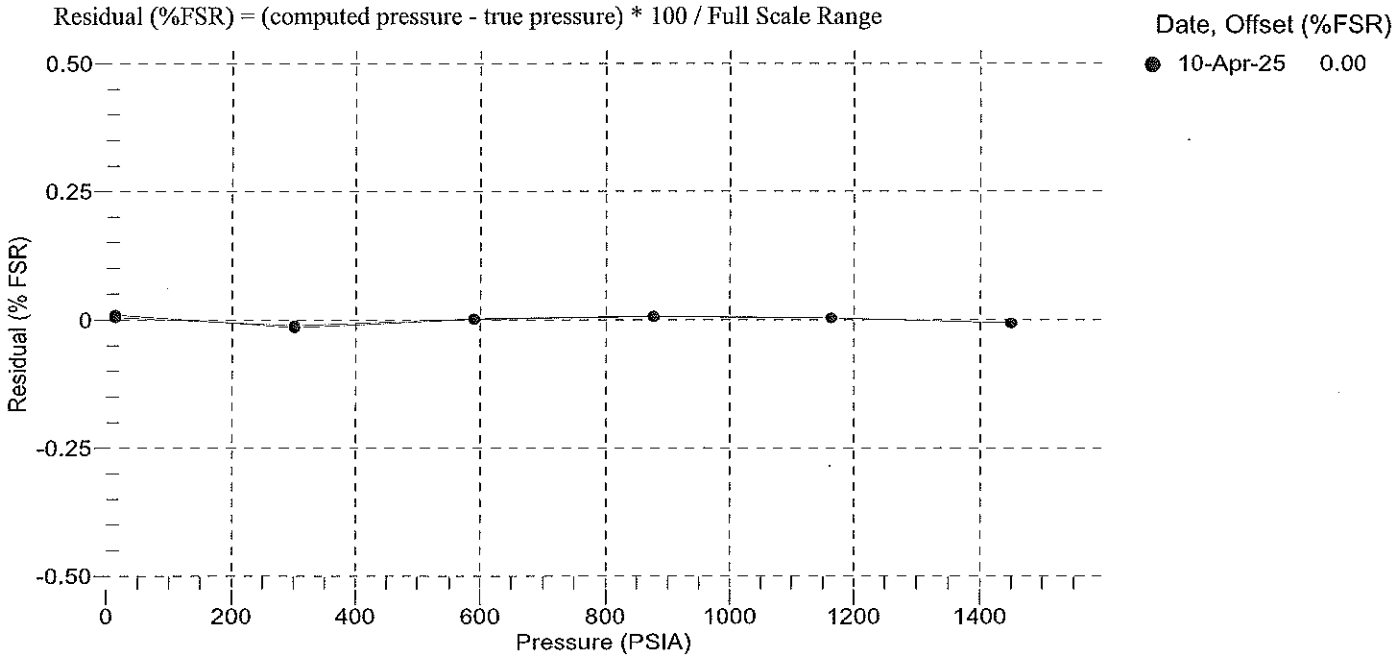
TEMPERATURE (°C)	SPAN
-5.00	25.12
35.00	25.16

y = thermistor output (counts)

$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range